

Graduate
Research & Arts
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Addressing Cold Start Problem in Knowledge Tracing for New Students via Few-Shot Model-Agnostic Meta Learning

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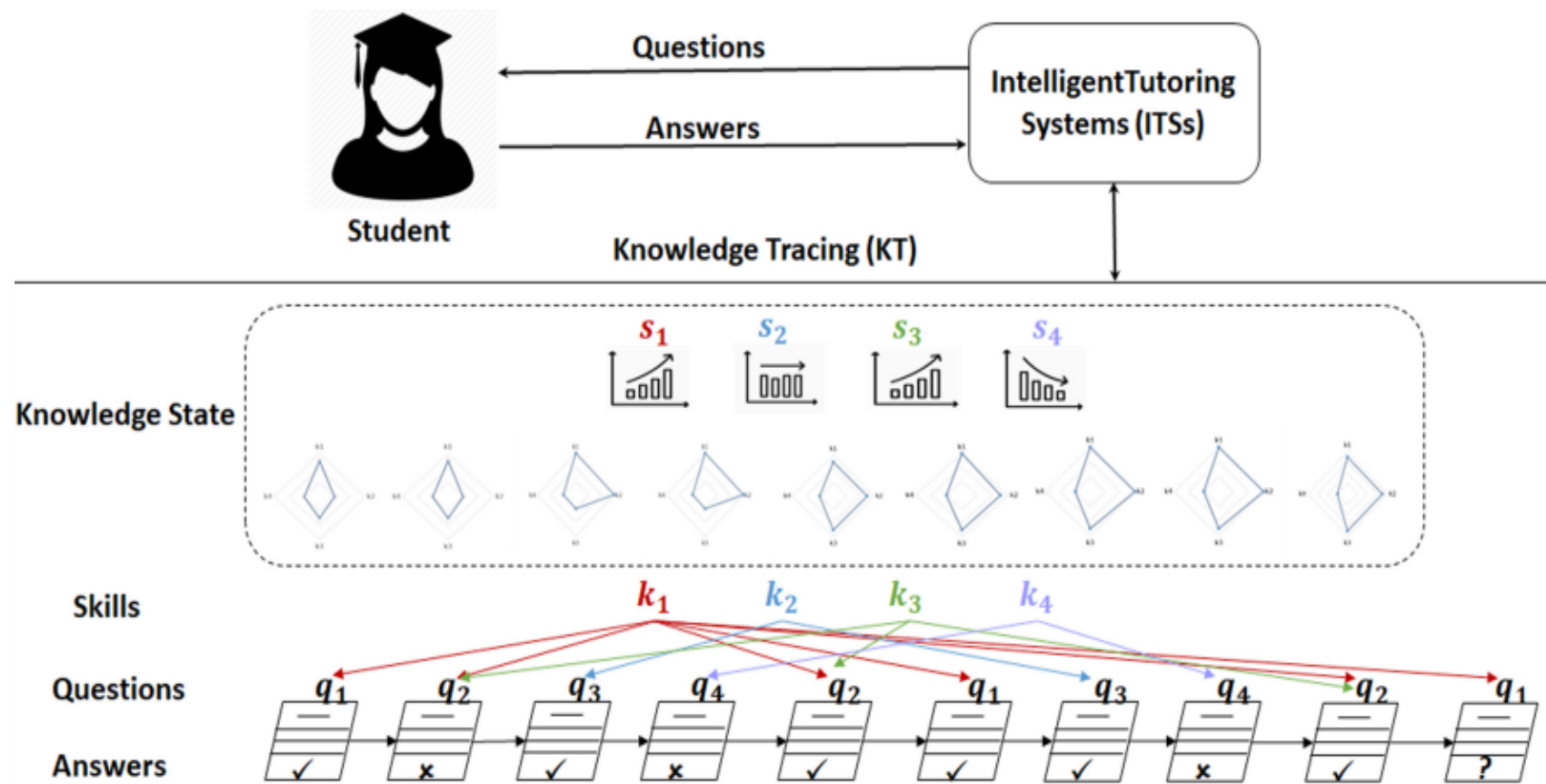


Introduction

- Cold Start problem in Knowledge Tracing refers to the difficulty Intelligent Tutor Systems face in accurately predicting a student's performance when they are new to a system
- We introduce a meta-learning approach designed to address the cold start problem where AI models struggle to predict the performance of new students with limited history
- Framing knowledge tracing as a few-shot learning problem
- A method that learns an optimal starting point for rapid personalization



Knowledge Tracing



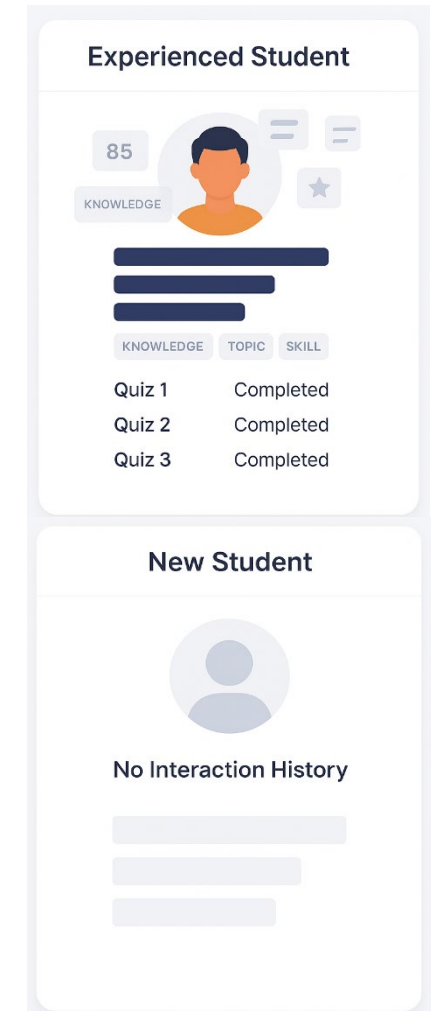
An example scenario for knowledge tracing in an Intelligent Tutoring System

Reference: Knowledge Tracing: A Survey, Ghodai Abdelrahman, Qing Wang, and Bernardo Nunes, ACM Computing Surveys, Vol. 55, No. 11, Article 22, 2023



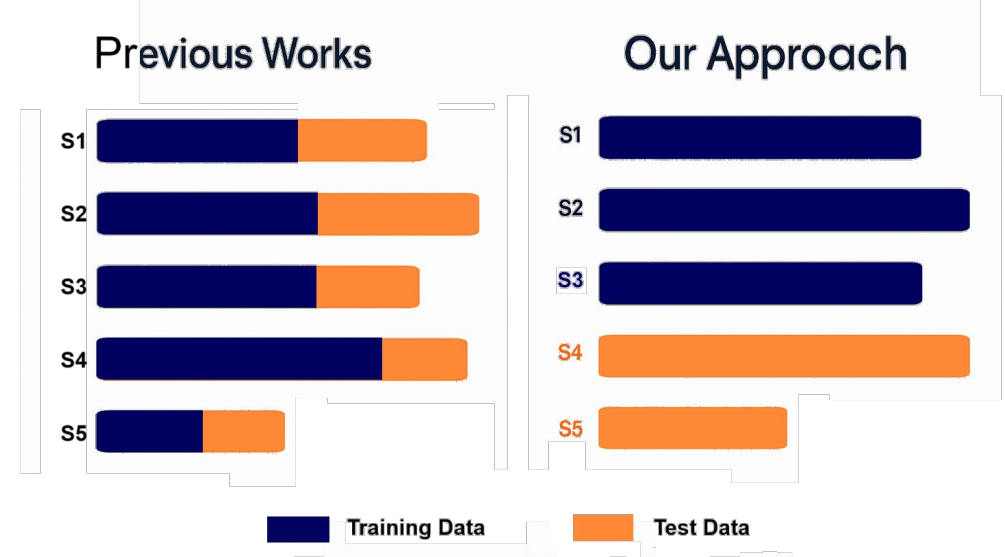
Understanding Cold Start in Knowledge Tracing

- A major challenge: **Cold Start Problem**
- Refers to
 - The challenge of accurately predicting a new student's knowledge state
 - When the system has **minimal or no prior interaction data** about them
- This delays effective personalization in Intelligent Tutoring Systems.
- Real-world systems often face this with brand-new learners logging in for the first time.



Cold Start Detection Approach

- Most KT models are still trained and tested on the same students
- Just different parts of their sequences.
- That doesn't tell us how well the model works for completely new learners
- Our study:
 - ✓ Trains on historical students.
 - ✓ Tests only on completely unseen learners.
 - ✓ Evaluation over multiple interactions.
- Provides a realistic cold start evaluation framework.



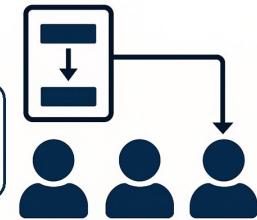
Test Set



Test Student



Training

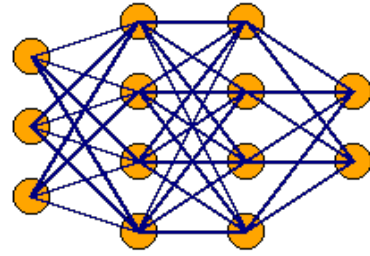


Students



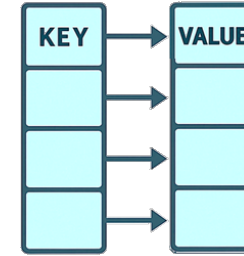
Models and Datasets Used

**3 Different
State-of-the-art
Models**



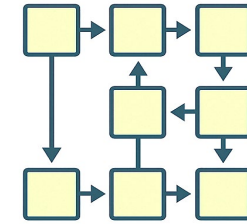
DKT

Deep Knowledge Tracing
(RNN Based)



DKVMN

Dynamic Key-Value Memory Network
(Memory Network Based)



SAKT

Self-Attentive Knowledge Tracing
(Transformer Based)

**3 Different
Concept-rich
Datasets**



ASSISTMENT 2009
(Large number of Skills)



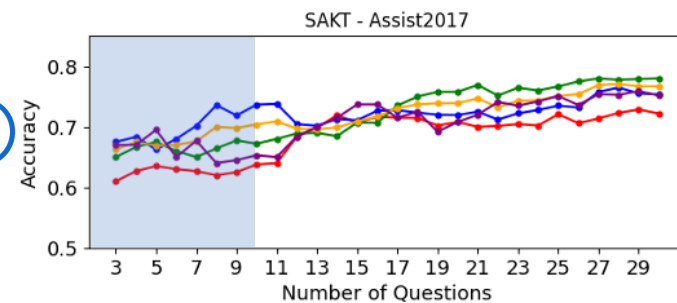
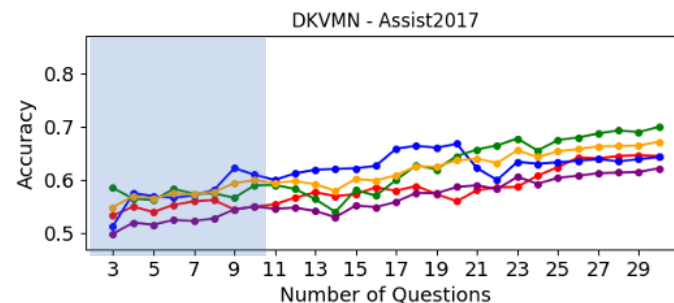
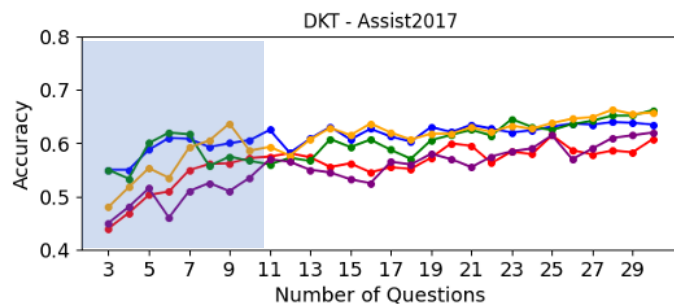
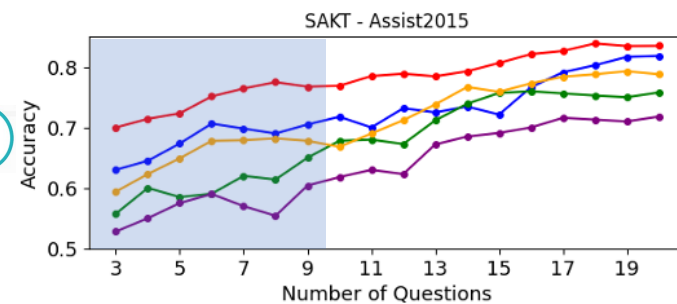
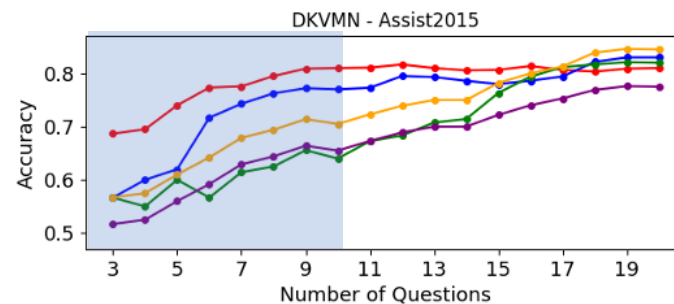
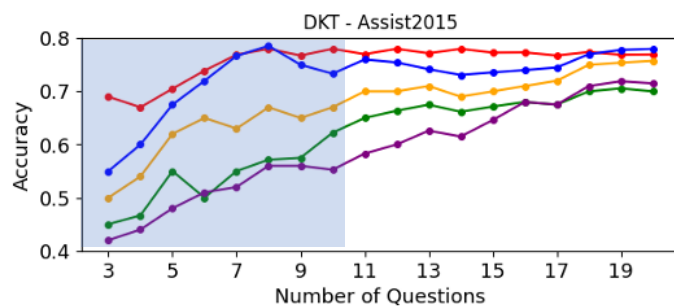
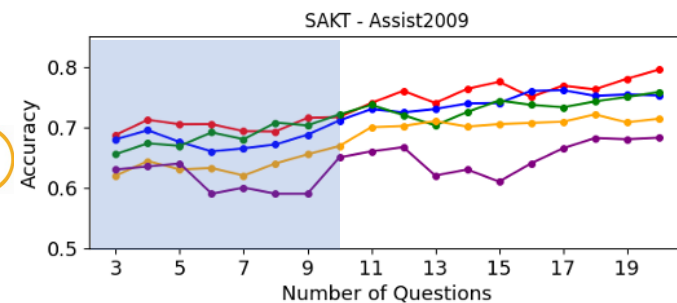
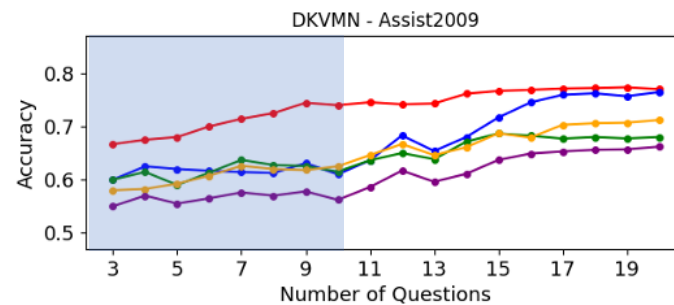
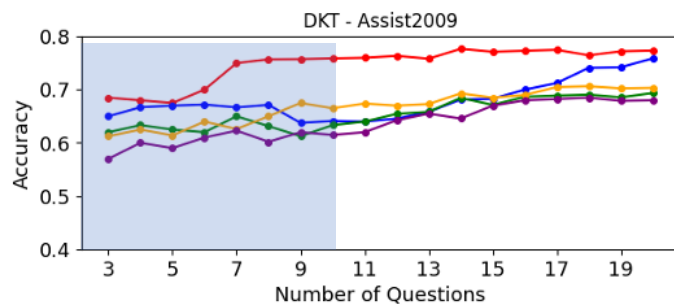
ASSISTMENT 2015
(Large Number of Students)



ASSISTMENT 2017
(Large Number of Interactions)



Cold Start Detection



Set1 Set2 Set3 Set4 Set5

Set1 Set2 Set3 Set4 Set5

Set1 Set2 Set3 Set4 Set5



Solution

MAML-KT: Model **A**gnostic **M**eta **L**earning for **K**nowledge **T**racing

- **Meta Learning**, often called "learning to learn", trains a model on many different tasks so it can recognize patterns and solve new, unseen problems with very little data.
- **MAML**, this algorithm identifies an ideal "starting point" for a model's settings, allowing it to reach peak performance after just one or two small updates.
- **MAML-KT** applies meta-learning to education by treating each student's unique learning journey as an individual task.
- Uses meta-learning to knowledge tracing by treating each student's unique learning journey as an individual task.



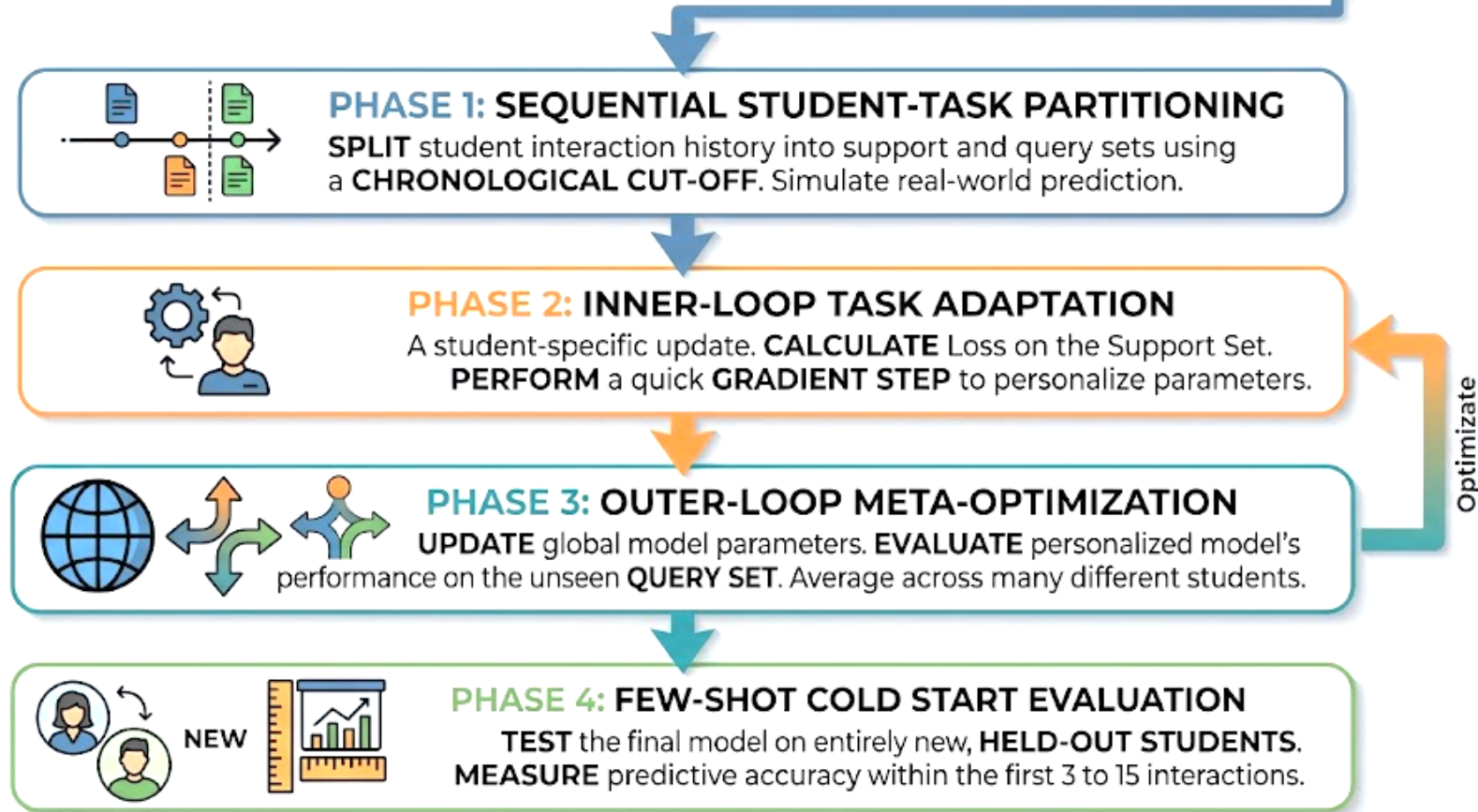
Research Questions

- 1.** Does meta learning improve early-phase new student performance over ERM baselines?
- 2.** Does the proposed MAML-KT approach scale with cohort size, i.e., do its cold start accuracy change when moving from the prior settings (10 students) to larger cohorts (20 and 50 students)?
- 3.** Under which sequence and content conditions does MAML-KT trail other ERM baselines? Specifically, how do context length and number and timing of skills first appearing in the query relate to the MAML-KT vs. others accuracy gap in the cold start range?



Methodology

METHODOLOGY FOR STUDENT-TASK PREDICTION



Algorithm

Algorithm 1: MAML-KT (GRU backbone, second-order)

Input: Train students $\mathcal{D}_{train}$, support size K , inner steps R , inner LR α , meta LR β , meta-batch B

- 1 **Preprocess:** For each student trajectory $\{(q_t, a_t)\}_{t=1}^T$, form next-step pairs (x_t, q_{t+1}, a_{t+1}) with $x_t = (q_t, a_t)$; discard $T < 2$ and enforce $1 \leq K \leq T - 1$.
- 2 **Model:** GRU over interaction tokens with projected target-item embedding.
- 3 **for** $epoch = 1, 2, \dots$ **do**
- 4 Sample meta-batch $\{(X_i, y_i, T_i)\}_{i=1}^B$ from $\mathcal{D}_{train}$;
- 5 $L_{meta} \leftarrow 0, V \leftarrow 0$;
- 6 **for** $i = 1$ **to** B **do**
- 7 **if** $T_i \leq K$ **continue**
- 8 **Causal split:**
- 9 support $(X^s, y^s) = (X_i[1:K], y_i[1:K])$;
- 10 query $(X^q, y^q) = (X_i[K+1:T_i - 1], y_i[K+1:T_i - 1])$;
- 11 $\phi \leftarrow \theta$; // fast parameters
- 12 **for** $r = 1$ **to** R **do**
- 13 $\ell_r \leftarrow \text{BCE}(f(X^s; \phi), y^s)$;
- 14 $\phi \leftarrow \phi - \alpha \nabla_{\phi} \ell_r$;
- 15 **end**
- 16 $\ell_q \leftarrow \text{BCE}(f(X^q; \phi), y^q)$;
- 17 $L_{meta} \leftarrow L_{meta} + \ell_q$; $V \leftarrow V + 1$;
- 18 **end**
- 19 **if** $V > 0$ **then**
- 20 $\theta \leftarrow \theta - \beta \nabla_{\theta} (L_{meta}/V)$;
- 21 **end**
- 22 **end**



Results

> Datasets

Assist 2009

Assist 2015

Assist 2017

> Cohort Size

10 20 50

> Cold Start Zone

Critical (First 10 Questions)

Moderate (11-15 Questions)

> Sets

5 for each zone

> Model Comparison

DKT (D)

DKVMN (K)

SAKT (S)

MAML-KT (M)

> Evaluation Metric

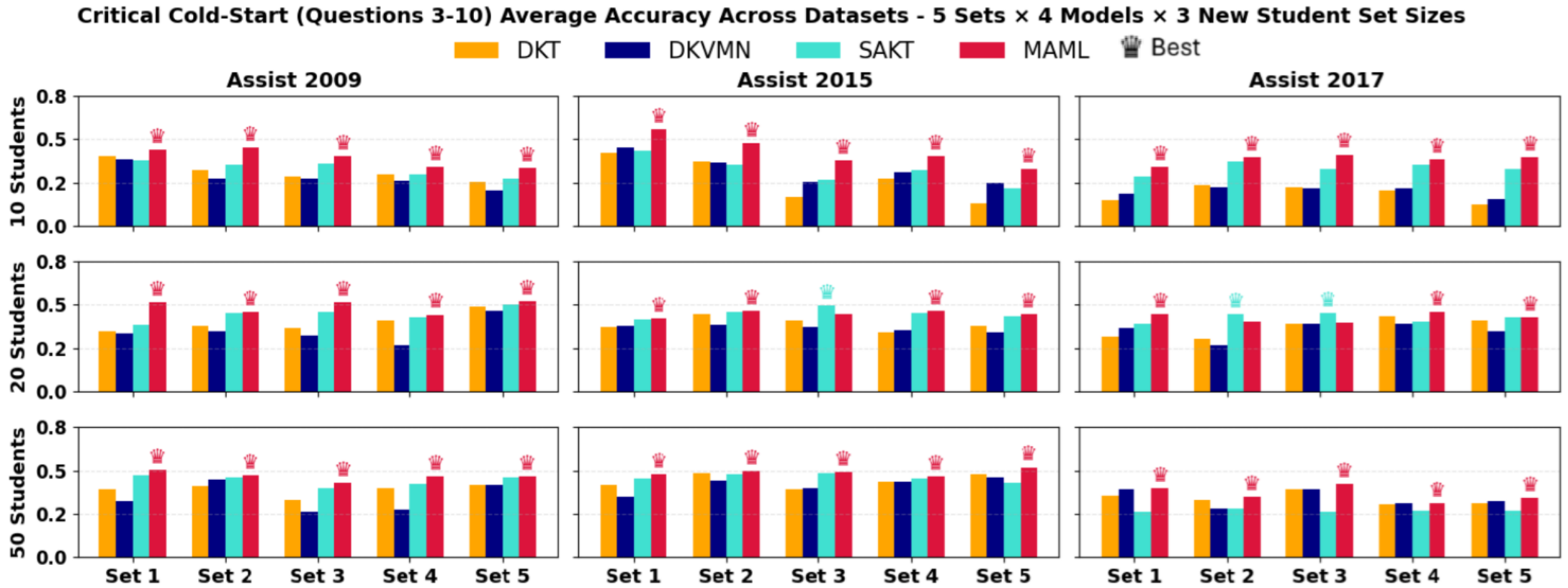
Average accuracy of prediction in that zone

Critical Cold Start						Moderate Cold Start				
Dataset	Set 1	Set 2	Set 3	Set 4	Set 5	Set 1	Set 2	Set 3	Set 4	Set 5
10 New Students										
ASSIST 2009	75.3 ^M	75.9 ^M	72.3 ^M	67.2 ^M	66.8 ^M	79.4 ^M	78.1 ^M	76.6 ^M	71.9 ^M	70.5 ^M
ASSIST 2015	84.5 ^M	78.1 ^M	70.0 ^M	72.1 ^M	66.0 ^M	88.3 ^M	78.7 ^M	76.8 ^M	77.9 ^M	75.6 ^M
ASSIST 2017	67.4 ^M	71.9 ^M	72.9 ^M	70.6 ^M	71.7 ^M	69.1 ^M	72.8 ^M	69.4 ^M	72.7 ^M	73.0 ^M
20 New Students										
ASSIST 2009	81.1 ^M	76.8 ^M	81.0 ^M	75.5 ^M	81.9 ^M	82.6 ^M	80.9 ^M	82.7 ^M	82.0 ^M	84.6 ^M
ASSIST 2015	73.6 ^M	77.3 ^M	79.5 ^S	77.3 ^M	76.0 ^M	76.8 ^M	80.7 ^M	78.0 ^M	81.7 ^M	80.1 ^M
ASSIST 2017	75.7 ^M	75.8 ^S	76.3 ^S	65.1 ^M	67.6 ^M	72.5 ^M	69.9 ^S	71.7 ^S	78.4 ^M	73.8 ^M
50 New Students										
ASSIST 2009	80.1 ^M	77.8 ^M	74.4 ^M	77.1 ^M	77.5 ^M	85.0 ^M	78.4 ^M	72.0 ^M	79.6 ^M	77.2 ^M
ASSIST 2015	78.2 ^M	79.9 ^M	79.3 ^M	77.2 ^M	81.5 ^M	79.7 ^M	81.8 ^S	82.6 ^M	80.1 ^M	80.2 ^M
ASSIST 2017	71.9 ^M	67.6 ^M	74.0 ^M	65.1 ^M	67.6 ^M	71.4 ^M	69.7 ^M	73.0 ^M	65.6 ^M	68.1 ^M

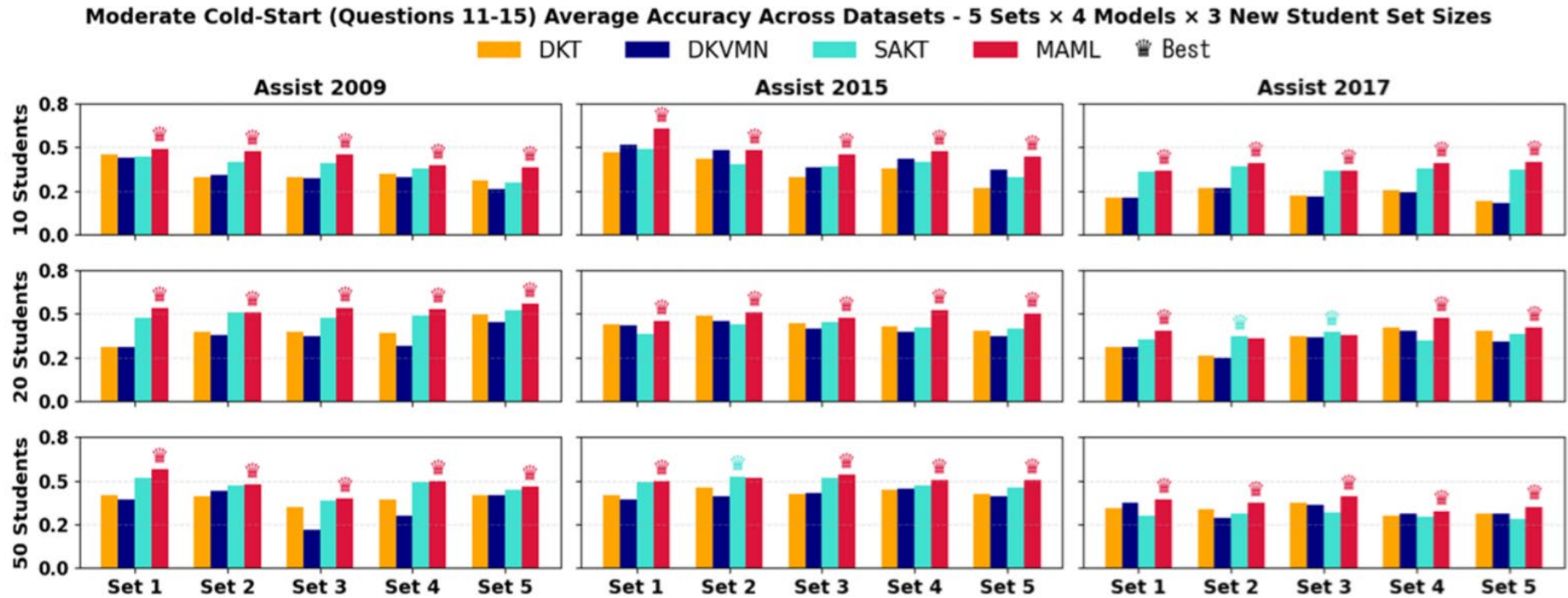
* M: MAML, D: DKT, N:DKVMN, S: SAKT



Results (Critical Cold Start)

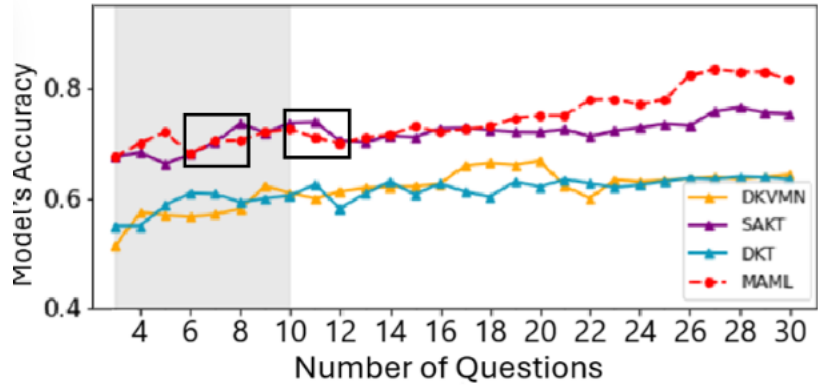


Results (Moderate Cold Start)



Results (Limitation)

(a) Model Accuracy vs Questions



(b) Per Student Answer Accuracy by Skill vs. Questions

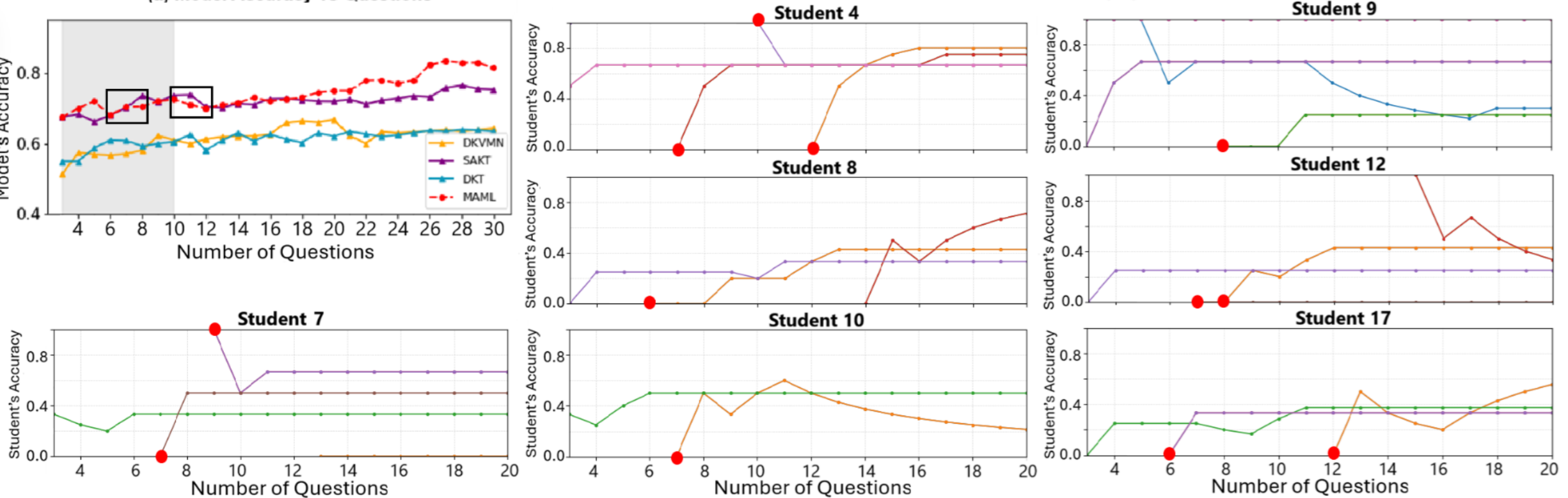


Fig. 4. Assist2017 - 20 New Students - Set 2, Questions 6-8 and 10-12 Explanation

(a) Model Accuracy vs Questions (b) Per Student Answer Accuracy by Skill vs Questions (The lines represent a specific skill and introduction of new skills in the Q6-8 and Q10-12 are marked with red circles)



Conclusion

- MAML-KT successfully addresses the cold-start problem by framing student knowledge tracing as a few-shot adaptation task from very limited interactions.
- The model significantly outperforms standard baselines in early-phase prediction accuracy
- Maintaining its effectiveness and showing that increasing the student cohort size also results in strong performance.
- Future research will focus on making adaptation strategies more robust for skill transitions and knowledge component changes.



Thank you!

For any more questions, please send me an email

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